Space Explanations of Neural Network Classification

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Abstract. We present a novel logic-based concept called *Space Explanations* for classifying neural networks that gives provable guarantees of the behavior of the network in continuous areas of the input feature space. To automatically generate space explanations, we leverage a range of flexible Craig interpolation algorithms and unsatisfiable core generation. Based on real-life case studies, ranging from small to medium to large size, we demonstrate that the generated explanations are more meaningful than those computed by state-of-the-art.

1 Introduction

Explainability of decision-making AI systems (XAI), and specifically neural networks (NNs), is a key requirement for deploying AI in sensitive areas [18]. A recent trend in explaining NNs is based on formal methods and logic, providing explanations for the decisions of machine learning systems [24, 30, 31, 40, 41, 43] accompanied by provable guarantees regarding their correctness. Yet, rigorous exploration of the *continuous* feature space requires to estimate *decision boundaries* with complex shapes. This, however, remains a challenge because existing explanations [24, 30, 31, 40, 41, 43] constrain only individual features and hence fail capturing *relationships* among the features that are essential to understand the reasons behind the multi-parametrized classification process.

We address the need to provide interpretations of NN systems that are as meaningful as possible using a novel concept of *Space Explanations*, delivered by a flexible symbolic reasoning framework where Craig interpolation [12] is at the heart of the machinery. When starting from a sample point, the explanation computation is quick, and yields generality and soundness of approximation. Interpolation extracts information from the proof of unsatisfiability which captures a reason why the given input cannot change classification. We showcase explanations for trained neural networks substantially more expressive and meaningful compared to prior work in formal XAI. For example, the classification of the heart attack risk is determined (among others) based on a non-trivial relation of age and cholesterol level of a patient, as illustrated in Figure 1. Our prototype

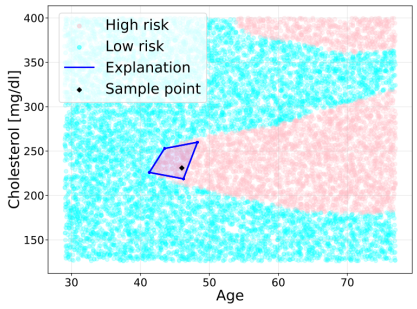


Fig. 1: Example of close approximations of non-trivial decision boundaries using meaningful explanations of the classification: a NN classifier of heart attack risk [25].

implementation of the framework, $\mathrm{S_P EX_{PL} AI_N}$, can automatically compute an explanation that closely approximates the shape of the decision boundary. Furthermore, such non-restricted explanations often cover many sample points at once.

This case study paper is an applied study using existing techniques (mainly Craig interpolation) in a new practical context (XAI). We share our lessons learned after experimenting with various real-life benchmarks, evaluate the quality of explanations generated using different interpolation techniques and further reduced using unsatisfiable cores. We demonstrate the benefits in carefully selected experiments with respect to the state-of-the-art to see how the techniques apply and how different strategies can be combined. For all kinds of benchmarks, no matter the size and the domain, the resulting explanations are more meaningful for the user. Moreover, computing interpolation-based explanations scales better with the size of the input than existing methods.

Related Work. Most of the classic approaches to XAI either perform analysis at the *unit (neuron) level* [17, 44], which however fails to capture global properties or interdependencies between units; are *gradient-based* methods [4, 42], which describe the network behavior around a particular sample; or *inversion* methods [29], which provide global views, but are still only approximating. There is also a range of *model-agnostic* methods [5, 28, 38], including SHAP [27] and LIME [37], which can be applied to any classifier (not only NNs), but often yield logically inconsistent explanations [31]. All [4, 5, 17, 27–29, 37, 38, 42, 44] rely on approximations or are *sample-based*, that is, the behavior outside the space of the underlying dataset or samples generated from the model remains unknown.

We focus on formal approaches to XAI that offer *strict guarantees*, exploiting computational engines like Marabou [26] or SMT solvers [6, 10, 11, 16, 22, 33]. State-of-the-art methods in this area address only special cases of our concept: constraints over individual features, leading to interval-based explanations [24] that cannot reflect feature relationships and hence approximate complex decision boundaries such as in Figure 1. Section 4 demonstrates generalizations of the

existing explanations, resulting in strictly larger spaces. Early methods in this domain aimed at so-called abductive explanations [23, 30, 31, 40, 41, 43] that are primarily sample-based, resulting in zero-volume spaces. The term *abduction* [1, 14, 35, 36] is misused, since unlike abductive explanations, it is not restricted to samples. All the approaches [23, 24, 30, 31, 40, 41, 43] also heavily depend on selecting a specific order of the features in which the constraints are relaxed. Furthermore, they might build an exponential number of verification queries to an off-the-shelf NN verification engine. In contrast, our method directly uses one satisfiability query to an interpolating SMT solver to generate an interpolant.

2 Background

A classification problem [20] is concerned with mapping input data into a predefined set of classes $\mathcal{K} = \{c_1, \ldots, c_K\}$. Given a set of *features* $\mathcal{F} = \{1, \ldots, m\}$, each $i \in \mathcal{F}$ takes values from a discrete or continuous *domain* $\mathcal{D}_i$. The *feature space* is defined as $\mathbb{F} = \mathcal{D}_1 \times \mathcal{D}_2 \times \cdots \times \mathcal{D}_m$. By $\mathbf{x} = (x_1, \ldots, x_m)$ we denote a point in $\mathbb{F}$, where each x_i is a *feature variable* (if clear from context, called features) taking values from $\mathcal{D}_i$. A *sample point* $\mathbf{s} = (s_1, \ldots, s_m) \in \mathbb{F}$ contains constants representing concretes value from $\mathcal{D}_i$, …, $\mathcal{D}_m$. A *classifier* $\mathcal{M} = (\mathcal{F}, \mathcal{D}, \mathbb{F}, \mathcal{K}, \kappa)$ contains a *classification function* $\kappa : \mathbb{F} \to \mathcal{K}$. An *instance* is a pair $(\mathbf{s}, c)$, where $\mathbf{s} \in \mathbb{F}$ and $c \in \mathcal{K}$ is a prediction $c = \kappa(\mathbf{s})$. Given a classifier $\mathcal{M} = (\mathcal{F}, \mathcal{D}, \mathbb{F}, \mathcal{K}, \kappa)$, the *class space* $\mathbb{F}_c \subseteq \mathbb{F}$ of $c \in \mathcal{K}$ is $\{\mathbf{x} \in \mathbb{F} \mid \kappa(\mathbf{x}) = c\}$. A *classification rule* for c is a formula φ_c such that $\forall\, \mathbf{x} \in \mathbb{F} \,.\, \varphi_c(\mathbf{x}) \iff \kappa(\mathbf{x}) = c$. If κ is defined on the whole feature space $\mathbb{F}$ then it can be partitioned as $\mathbb{F} = \bigcup_{c \in \mathcal{K}} \mathbb{F}_c$, and for all $a, b \in \mathcal{K}$, if $a \neq b$ then $\mathbb{F}_a \cap \mathbb{F}_b = \emptyset$ (because κ must not be ambiguous). A *class boundary* of c is the topological boundary of $\mathbb{F}_c$. The *decision boundary* is the union of class boundaries of all classes $c \in \mathcal{K}$ in a classifier $(\mathcal{F}, \mathcal{D}, \mathbb{F}, \mathcal{K}, \kappa)$.

3 Space Explanations

Explanations produced by existing tools do not capture relations between the input features, and consequently, they cannot approximate the decision boundary. To address it, this paper presents a *logic*-based approach to compute explanations with the flexibility of representing different shapes of the feature space.

Definition 1 (Space Explanation, Impact Space). *Given a classifier $\mathcal{M}$ computing a classification function κ from feature space $\mathbb{F}$, and a class c, a* space explanation *of c is a logic formula φ such that $\forall\, \mathbf{x} \in \mathbb{F} \,.\, \varphi(\mathbf{x}) \implies \kappa(\mathbf{x}) = c$. The* impact space *of φ, $\mathbb{F}_\varphi \subseteq \mathbb{F}_c$, is a set $\{\mathbf{x} \in \mathbb{F} \mid \varphi(\mathbf{x})\}$.*

Space explanations represent sufficient conditions of classification to class c. Since $\varphi \implies \varphi_c$, classification rules are also space explanations. Hence, we have a general concept of explanations with the following benefits:

1. The *shape* of the explanations is *not restricted*, hence it is possible to approximate arbitrarily complex class spaces and decision boundaries.

2. It is possible to *capture relationships* among features and truly *explain* the *reasons* behind the classification.
3. Space explanations can be merged or intersected with each other.

In Figure 1, for example, items 1 and 2 addressed by accurate approximation of the decision boundary using an explanation in the shape of a convex polygon.

We assume that for all classes, at least one sample point is classified into the class, meaning that none of the classes are redundant. Given a formula $\psi := \psi_{\mathcal{M}} \wedge \psi_{\mathcal{D}} \wedge \neg\psi_c$ where: $\psi_{\mathcal{M}}$ encodes the neural network $\mathcal{M}$, $\psi_{\mathcal{D}}$ encodes the domains $\mathcal{D}$ of the feature space, i.e. $\mathbf{x} \in \mathbb{F}$, $\neg\psi_c$ encodes the constraint that the outcome of the classification is *not* class c, it is *satisfiable* with no additional restrictions. Now suppose a sample point $\mathbf{s}$ classified as class c. Using the encoding $\varphi_{\mathbf{s}} := \bigwedge_{i \in \mathcal{F}} x_i = s_i$, formula $\varphi_{\mathbf{s}} \wedge \psi$ is in turn *unsatisfiable*.

This principle can be generalized by exploiting the fact that a space explanation φ of class c *guarantees* classification to the class for all points covered by the explanation, so formula $\varphi \wedge \psi$ is also unsatisfiable. Such a formula enables the use of Craig interpolation[5]: interpolant $I = \mathit{Itp}(\varphi, \psi)$ is a space explanation of c and it satisfies $\varphi \implies I$ (and hence $\mathbb{F}_\varphi \subseteq \mathbb{F}_I$).

Hence, we have a universal means of generalization of existing space explanations, still guaranteeing the classification. Although space explanations are not tied to specific data samples, formulas $\varphi_{\mathbf{s}}$ can be used as a starting space explanation, for which the interpolants are computed quickly because all input variables are fixed to concrete values. This offers the following improvements over existing methods:

1. The captured feature relationships stem from a mathematical *proof* of the classification, hence providing *meaningful* information.
2. The concept is *flexible*, offering to use *arbitrary* Craig interpolation algorithms. The resulting explanations exhibit various logical strengths and forms.
3. It is possible to further *generalize* any *existing* space explanation.

Interpolation algorithms can benefit from special cases [19] where a formula can be partitioned. We address just one example when a space explanation can be split into two parts $\varphi_A \wedge \varphi_B$: given a Craig interpolation procedure *Itp* with $I = \mathit{Itp}(\varphi_A, \varphi_B \wedge \psi)$, then $I \wedge \varphi_B$ is a logically weaker space explanation. We used this technique to identify pair feature relationships in Figure 1.

Figure 2 gives a high-level representation of a fully functioning and flexible algorithmic *framework* that provides expansion of space explanations using a collection of *strategies* listed below.

– **Generalize (G).** Compute Craig interpolants on an arbitrary space explanation. We use the algorithms based on Farkas' lemma (Itp_F) or their logically stronger *decomposing* variants [7] (Itp_{DF}), the dual versions of these algorithms (Itp'_F, Itp'_{DF}) to yield weaker interpolants [15], and an algorithm [3]

[5] Given an unsatisfiable formula $A \wedge B$, a *Craig interpolant* [12] is a formula I such that A implies I, $I \wedge B$ is unsatisfiable, and I uses only the common variables of A and B. We denote interpolant I computed by an interpolation procedure *Itp* from formulas A and B by $I = \mathit{Itp}(A, B)$.

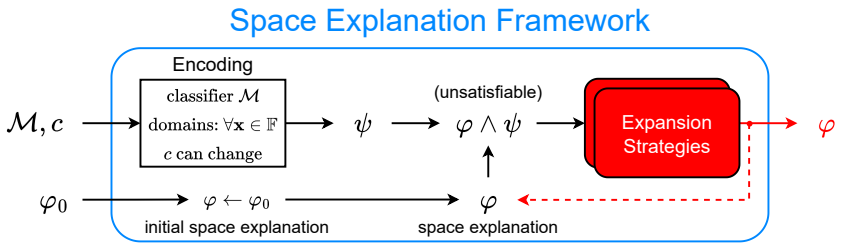


Fig. 2: Algorithmic framework: starting from an initial space explanation φ_0, the impact space is expanded and a logically weaker explanation φ is produced. The user may select from multiple strategies for further optimization (cf. the dashed arrow).

(Itp_f) parametrized by a rational factor $f \in [0, 1]$ with the logical strength in between Itp_F and Itp'_F. These arithmetic interpolation algorithms are combined with McMillan's propositional interpolation algorithm [32].
- **Reduce (R).** Weaken the formula and reduce size by computing an unsatisfiable core. To get irreducible explanations ($\mathbf{R}_{min}$), we use exhaustive minimization which may introduce significant overhead.
- **Capture (C).** Partition an interval-like formula (e.g. a sample) and **Generalize**. Weaken only part φ_A with constraints over at least one of the given features, hence capturing their mutual relationships.

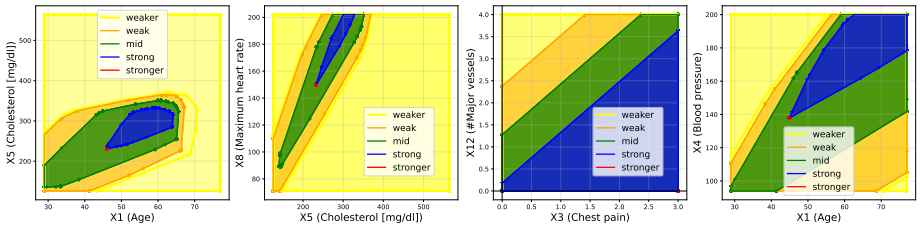
Strategy **Generalize** and **Capture** is often followed by **Reduce** to simplify the formulas from unnecessary constraints.

4 Experimental Evaluation

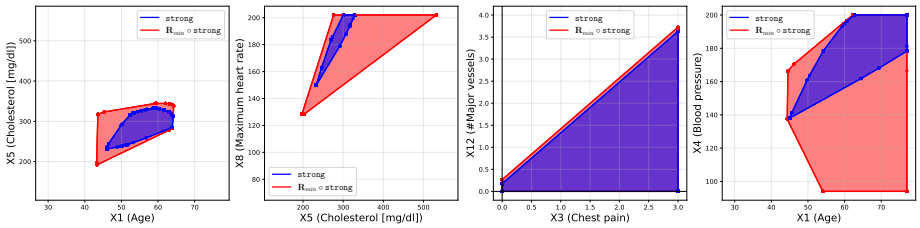
We showcase[6] the computation of space explanations for selected trained neural networks and demonstrate that they are substantially more expressive and meaningful since our approach captures feature relationships and approximates decision boundaries. We implemented the novel Space Explanation Framework in a prototype tool $\mathrm{S_PEX_{PL}AI_N}$[7], focusing on QF_LRA logic, on top of the OpenSMT2 solver [8, 22] which is an interpolating solver that comes with a set of techniques assembled into the integral framework, combining SMT solving with the computation of the Craig interpolants of various size and strength [2, 3, 7, 9, 21, 39]. We evaluated the following NN datasets and models:

- Heart attack dataset [25] focuses on predicting the risk of heart attacks based on various medical indicators of patients, it contains 13 input features and 2 possible classification outcomes: high or low risk. We trained the NN using one hidden layer with 50 neurons, and used a dataset with 303 sample points.

[6] All table contents are reproducible: https://doi.org/10.5281/zenodo.15490124
[7] https://github.com/usi-verification-and-security/spexplain



(a) Generalize (G): Interpolation-based explanation impact spaces (projections)



(b) Reduce ($\mathbf{R}_{min} \circ \mathbf{G}$): **G-stronger** explanations and their exhaustive simplifications (projections)

Fig. 3: Projections into selected pairs of features of interpolation-based explanations (**G**) of the heart-attack risk model, possibly combined with reduction ($\mathbf{R}_{min}$).

– Obesity dataset [34] provides data for estimating obesity levels in individuals based on their eating habits and physical condition, resulting in 15 input features and 7 classes. We trained the model using 3 hidden layers with 10, 20, and 10 neurons, respectively, and used a dataset with 50 sample points.
– MNIST dataset [13] is We trained the model using 784 inputs and a hidden layer with 200 neurons, and used a dataset with 50 sample points.

Instances of interpolation algorithms use the notation $Itp_{DF} \mapsto$ `stronger`, $Itp_F \mapsto$ `strong`, $Itp'_F \mapsto$ `weak`, $Itp'_{DF} \mapsto$ `weaker`, and $Itp_f \mapsto$ `mid` with $f := 0.5$. The evaluation uses 2 hour *time-out* to compute all explanations for a dataset and an explanation setup, and was performed on a Linux 5.4 machine with 256 GB physical memory and AMD® EPYC 7452 32-core CPU.

Evaluation of Strategies. Instantiations of strategy **Generalize** (**G**) provide flexibility of covering spaces with various size, illustrated in Figure 3a on selected sample points from the dataset. We resolve the issue of visualizi ng high-dimensional spaces by selecting candidate pairs of features from the heart attack dataset and plotting the projection[8] of the impact spaces into two dimensions. The examples reveal the similarity of feature relations despite aiming at different logical strengths. However, we spot that `weaker` here always covers the whole

[8] Projections offer a visual intuitive understanding of the geometrical differences between impact spaces. While not capturing the complete information contained in higher-dimensional spaces, they are still more informative than simple slices.

Table 1: Average performance of strategy **G** using different *Itp* algorithms.

	Heart attack			Obesity			MNIST		
Itp algorithm	relaxed	#terms	time[s]	relaxed	#terms	time[s]	relaxed	#terms	time[s]
`stronger`	0%	20.1	0.03	0%	29.0	0.30	0%	927.3	9.17
`strong`	97%	51.0	0.04	72%	45.8	0.30	100%	209.0	10.12
`mid`	100%	51.0	0.04	100%	45.8	0.34	100%	209.0	10.25
`weak`	100%	51.0	0.04	100%	46.1	0.30	100%	209.0	10.42
`weaker`	100%	198.2	0.04	100%	64.1	0.42	100%	67330.6	10.55

projection[9], while `stronger` in turn results just in the original sample point. This is no exception: Table 1 reports that `stronger` relaxes no features at all (i.e. each feature is still fixed to the original value), producing way too strong interpolants. Algorithm `strong` sometimes does not relax all features. Next, the table reports the average size of the resulting formulas term-wise (the number of in/equalities). The average runtimes per sample point are rather small and do not vary among the approaches, but are sensitive to more complex models.

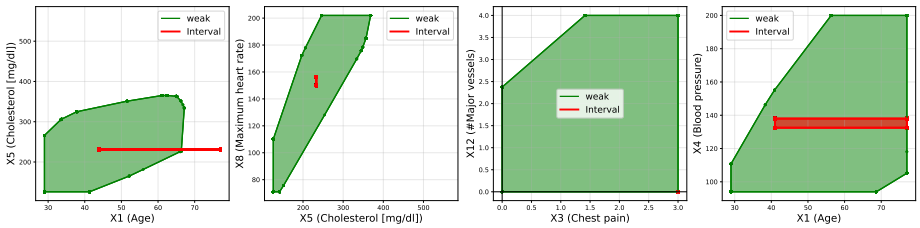
Strategy **Reduce** (**R**, $\mathbf{R}_{min}$) introduces a trade-off between formula simplification and runtime overhead when applied on top of **Generalize** (e.g., $\mathbf{R} \circ \mathbf{G}$), using an unsatisfiable core (**R**) or an irreducible one ($\mathbf{R}_{min}$). Figure 3a demonstrates side by side with Figure 3b that the projection of `strong` is further expanded by $\mathbf{R}_{min}$ (**R** expands none of the four presented `strong` projections), but in different directions compared to weaker algorithms. We confirmed this phenomenon by running extensive subset-comparisons which in cases such as $(\texttt{strong}, \mathbf{R}_{min})$ vs. $(\texttt{weak}, -)$ often yielded not comparable (**NC**) results: the spaces intersect but none of them subsumes the other. Table 2 shows that $\mathbf{R}_{min}$ significantly reduces the formula but with a high cost, while **R** offers a convenient balance. The reduction times-out (X) with more complex models.

Figure 4a follows with a comparison of projections between strategy **G** (cf. Figure 3a), specifically selecting `weak` algorithm that produces large yet convex impact spaces, and an approach that is based on intervals, denoted as **I**, that computes the explanations on top of an irreducible abductive explanation, denoted by **A**, resulting in $\mathbf{I} \circ \mathbf{A}$. Since the state-of-the-art-approach [24] is implemented for decision trees and not for NNs, we implemented the specialized computation of interval explanations using the Marabou verifier and a strict limit on the number of attempts of particular relaxations. We concentrate on comparison using the heart attack model. The projections of interval-based explanations form rectangles, lines, or even single points due to the lack of their expressivity. In contrast, **G** explanations cover larger areas with less limited shapes, yet do not entirely subsume interval explanations: Even if a projection is subsumed, it does not mean that the space is subsumed in the other dimensions as well. While the intervals are being relaxed in a certain order, hence strictly

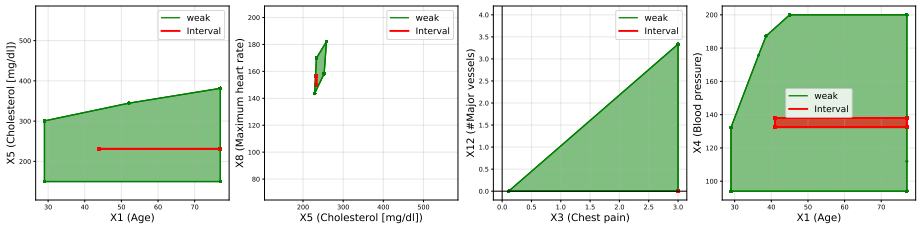
[9] Notably, the explanation does not cover the whole feature space. However, high-dimensional projections (e.g., from 13D to 2D) collapse information from other dimensions, making explanations appear larger.

Table 2: Time overhead and impact of **Reduce** on top of **Generalize** ($\mathbf{R} \circ \mathbf{G}$).

	G	#terms			time [s]		
	Itp algorithm	–	**R**	$\mathbf{R}_{min}$	–	**R**	$\mathbf{R}_{min}$
Heart attack	`stronger`	20.1	20.1	9.5	0.03	0.03	3.14
	`strong`	51.0	44.6	3.8	0.04	0.83	16.77
	`mid`	51.0	39.4	4.6	0.04	0.66	11.80
	`weak`	51.0	38.8	6.4	0.04	0.56	7.95
	`weaker`	198.2	197.9	25.9	0.04	0.08	7.14
Obesity	`stronger`	29.0	29.0	X	0.30	0.38	X
	`strong`	45.8	41.7	X	0.30	7.94	X
	`mid`	45.8	42.7	X	0.34	20.62	X
	`weak`	46.1	41.7	X	0.30	35.37	X
	`weaker`	64.1	58.0	X	0.42	23.44	X
MNIST	`stronger`	927.3	927.3	X	9.17	10.45	X
	`strong`	209.0	X	X	10.12	X	X
	`mid`	209.0	X	X	10.25	X	X
	`weak`	209.0	X	X	10.42	X	X
	`weaker`	67330.6	X	X	10.55	X	X



(a) Projections of **G** explanations (green) compared to $\mathbf{I} \circ \mathbf{A}$ explanations (red)



(b) Projections of **G** explanations (green) on top of $\mathbf{I} \circ \mathbf{A}$ explanations (red), resulting in $\mathbf{G} \circ \mathbf{I} \circ \mathbf{A}$

Fig. 4: Comparing projections of interpolation-based (**Generalize**, **G**) and interval explanations ($\mathbf{I} \circ \mathbf{A}$) into selected pairs of features of the heart-attack risk model.

preferring certain directions over others, the interpolation aims at more general expansions. Furthermore, the expansion of the interval explanations is limited by decision boundaries in the other dimensions. This observation is confirmed by pairwise subset-comparisons, consistently arriving at **NC** results.

Table 3: Average performance of **G** vs. **A** and **I** strategies in heart-attack model.

	relaxed	#terms	#solver calls	time [s]
A	38%	8.1	13	0.08
I ∘ **A**	79%	9.3	40.4	0.53
G	100%	51.0	1	0.04
G ∘ **A**	100%	45.3	1	0.39
G ∘ **I** ∘ **A**	100%	63.5	1	2.53

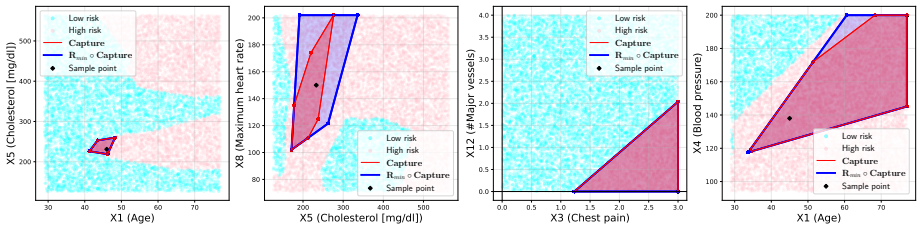


Fig. 5: Approximation of decision boundaries and feature relations using strategies **C** and $\mathbf{R}_{min} \circ \mathbf{C}$ within selected pairs of features of the heart-attack risk model.

Table 3 gives comparison of the average performance of the approaches (similarly as in Table 1), including irreducible abductive explanations (**A**). **G** explanations (row #3) are computed even faster than **A** explanations, requiring just one call to the logical solver. Interval explanations did not always succeed in relaxing all features. Nevertheless, the formulas are simpler when based on intervals or samples. Formulas from **G** could be reduced but would exceed the computation time of intervals. The observations remain similar when using other algorithms than `weak`. Figure 4b further illustrates the potentials of strategy **G** when running on top of arbitrary existing explanations, such as interval explanations (i.e., **G** ∘ **I** ∘ **A**), yielding strictly larger spaces (confirmed by subset-comparison checks). Nonetheless, the projections of the interpolants in Figures 4a vs. 4b (i.e., **G** vs. **G** ∘ **I** ∘ **A**) sometimes differ due to more guided constraints induced by the intervals. Next, we revisit Table 3 on rows #4–5, including **G** ∘ **A** as well. The observations remain similar except that when generalizing on top of a more general starting point, the runtime increases. Still, our generalization of **A** explanations runs faster than if using intervals (i.e., **I** ∘ **A** vs. **G** ∘ **A**).

Strategy **Capture** (**C**) directly aims at identifying relationships among selected features and consequently at approximating decision boundaries. The generalization is guided because it focuses on selected dimensions and picks a slice of the feature space, leaving all other features fixed to the original values. Although such limited exploration does not reveal anything about other dimensions, it extracts partial information helpful to understand the classification. Moreover, it is easy to sample two or three-dimensional slices and compare the explanations with estimated class spaces and decision boundaries. Figure 5 shows the explanations of the sample points and feature pairs as in Figures 3 and 4, this time

Table 4: Quantitative subset-comparison of **C** vs. **G** explanations, combined with **R** or $\mathbf{R}_{min}$, ranging over selected features where the others are fixed to the original values.

Features	**C**	**G** (sliced)	$\supset$	$=$	$\subset$	**NC**
x_1, x_5	$\mathbf{R} \circ \mathbf{C}$ vs.	$\mathbf{R} \circ \mathbf{G}$	8%	43%	41%	7%
	$\mathbf{R}_{min} \circ \mathbf{C}$ vs.	$\mathbf{R}_{min} \circ \mathbf{G}$	65%	0%	0%	35%
x_3, x_{12}	$\mathbf{R} \circ \mathbf{C}$ vs.	$\mathbf{R} \circ \mathbf{G}$	2%	92%	5%	1%
	$\mathbf{R}_{min} \circ \mathbf{C}$ vs.	$\mathbf{R}_{min} \circ \mathbf{G}$	89%	5%	0%	6%
x_1, x_4	$\mathbf{R} \circ \mathbf{C}$ vs.	$\mathbf{R} \circ \mathbf{G}$	9%	54%	31%	6%
	$\mathbf{R}_{min} \circ \mathbf{C}$ vs.	$\mathbf{R}_{min} \circ \mathbf{G}$	84%	0%	0%	16%

Table 5: Average performance of strategies **C** and **G** from Table 4 possibly in combination with **Reduce** (**R** or $\mathbf{R}_{min}$).

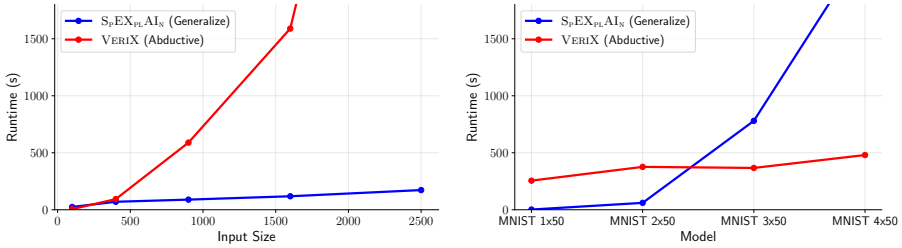
	Heart attack				Obesity				MNIST			
	C		**G** (sliced)		**C**		**G** (sliced)		**C**		**G** (sliced)	
Reduce	#terms	time[s]	#terms	time[s]	#terms	time[s]	#terms	time[s]	#terms	time[s]	#terms	time[s]
–	61.2	0.04	62.0	0.04	59.1	0.33	59.1	0.31	938.9	9.65	991.0	10.54
R	47.0	0.09	49.8	0.56	41.9	1.15	54.7	35.3	834.7	47.1	X	X
$\mathbf{R}_{min}$	9.4	2.52	17.4	7.94	X	X	X	X	X	X	X	X

using **C** instead of **G**, sticking to algorithm `weak`. The interpolation captures even non-trivial decision boundaries and some of the relations intuitively resemble real-world phenomena. For example, the plot on the right identifies a high risk of heart attack according to increasing age and decreasing blood pressure.

Using **Reduce** (**R**, $\mathbf{R}_{min}$) is especially useful for explanations produced by strategy **C** to enhance their interpretability. Furthermore, the reduction is often more efficient than when applied to **G**. Table 4 shows the quantitative subset-comparison, that is, how many explanations in percentage exhibited the relation superset ($\supset$), equivalent ($=$), subset ($\subset$), or not comparable (**NC**), between **C** and **G** explanations that have been reduced and while focusing just on the slice of selected pairs of features (cf. Figures 3–5). With no reduction, the slices are equivalent. Yet, the formulas differ, because strategy **C** separates the fixed features from the focused ones[10]. Consequently, the opportunities for reductions offered by the structure of the formulas are different. While strategy **R** usually expands the explanations better when applied to **G** (i.e. $\mathbf{R} \circ \mathbf{G}$), $\mathbf{R}_{min} \circ \mathbf{C}$ always produced a larger space within the slice than $\mathbf{R}_{min} \circ \mathbf{G}$, if excluding **NC** cases. The results for x_5, x_8 are almost the same as for x_1, x_5. Finally, Table 5 shows that the simplification is more efficient in terms of runtime and #terms when using the focused strategy **C** than if interpolating the whole formula with **G**.

Scalability. We conducted two experiments to evaluate how our prototype tool $\mathrm{S_PEX_{PL}AI_N}$ scales with increasing input size and depth of the NN using MNIST. We compare our method with VeriX [43], a state-of-the-art abductive explana-

[10] Example: **G** sliced to x_1, x_2 yields $\big(2x_1 + x_3 \geq 7 \wedge x_1 - x_2 + x_3 \geq 2\big) \wedge x_3 = 1$ which is equivalent to $\big(x_1 \geq 3 \wedge x_1 - x_2 \geq 1\big) \wedge x_3 = 1$ produced by **C** of x_1, x_2.



(a) Scalability with input size **(b)** Scalability with NN depth

Fig. 6: Scalability comparison of $S_PEX_{PL}AI_N$ and VeriX on MNIST

tion algorithm[11]. While the approaches generate different types of explanations and are not directly comparable, their relative scalability can still be compared.

The first experiment shows how the runtime grows with input size. For that, we used a fixed NN architecture with two hidden layers of size 50[12], and resized MNIST images to input dimensions 10×10, 20×20, 30×30, 40×40, and 50×50. Figure 6a shows how the runtime grows with the input size for computing **G** explanations and abductive explanations: the runtime of VeriX increases steeply, while **Generalize** remains relatively stable. The difference is caused by the fact that VeriX performs one verification query per feature, while our method requires only a single query to generate an explanation.

The next experiment examines scalability with respect to model depth. We trained NNs with 1 to 4 hidden layers, each having 50 neurons. Figure 6b shows that the runtime of VeriX grows moderately with the number of layers, while the runtime of **Generalize** increases more significantly with deeper networks. This is likely because, unlike VeriX, our prototype tool does not apply any NN-specific optimizations within its underlying constraint solver. Addressing this limitation is left for future work.

Lessons Learned. The following table summarizes our observations regarding the computation, interpretation, simplification, and comparison of explanations.

Observations	**Solution**
Problem: Explanations produced by **Generalize** are too complicated to read.	
1. `simplify` command of Z3 or CVC5 SMT solvers is insufficient. 2. **Reduce** simplifies conjunctive formulas, often surprisingly well. 3. McMillan interpolation algorithm extends formulas into conjunctions and its dual into disjunctions.	Use the non-dual algorithm and simplify the explanations using **Reduce** if needed.
Problem: Information regarding just a few features is difficult to extract, even if we use **Reduce**.	
1. Partitioning interval-like formulas separates selected features.	Use **Capture** and **Reduce**.

[11] https://github.com/NeuralNetworkVerification/VeriX, commit b6b2cc0

[12] A smaller network was used to reduce the number of time-outs.

Problem: It is not clear which interpolation algorithm to choose.	
1. The yielded formulas differ in logical strength and length. 2. The weaker the formula, the larger the space covered, offering the best opportunity to e.g. approximate decision boundaries. 3. Smaller or more focused formulas might be easier to interpret.	Use multiple algorithms depending on the current application and flexibly pick suitable outcomes.

Problem: It is not clear how to visualize explanations with high-dimensional impact spaces.	
1. Projections (cf. Figure 3) may yield too large area since the information from the other dimensions is collapsed. 2. Slices fix all other dimensions (cf. Figure 5), but conclusions on those dimensions are very limited. However, they allow direct comparison with decision boundaries or their sampling.	Use slices to show local information and to compare with decision boundaries. Use projections to compare the robustness of explanations.

Problem: Sometimes, quantitative and more rigorous comparisons than visualization are needed.	
1. Computing the volume of high-dimensional spaces is non-trivial. 2. It is easy to check if a space explanation is subsumed by another using the implication of the formulas (in conjunction with $\psi_{\mathcal{D}}$). 3. Spaces often intersect but are not entirely subsumed by another.	Estimate the quality using multiple metrics. If needed, include a visual comparison.

Idea: What if we compared only selected features?	
1. Computing the volume of 2D or 3D spaces is feasible. 2. Quantifier elimination yields exact projections but is expensive. 3. Projections can be approximated via linear programming. 4. Whole explanations can be compared by exhaustively comparing their projections into all particular pairs (or triplets) of features.	Comparing only selected features is a simpler problem and may also enable more thorough comparisons.

5 Conclusion and Future Work

This paper presented a novel concept of provably correct, logic-based space explanations of the classification process of neural networks. The explanations are associated with complex-shaped spaces and capture relations among the features that stem from mathematical proofs, substantially improving the approximation of decision boundaries over existing methods. The Space Explanations concept is supported by a flexible framework of algorithms including efficient Craig interpolation-based techniques and unsatisfiable core extraction to compute an extensive range of different yet meaningful explanations. On real-world neural networks trained on practical datasets, we performed a series of experiments and computed explanations of different quality. The evaluation of our case studies confirms that our algorithms yield explanations that are more general than existing explanations. We shared lessons learned during the extensive experimentation with the new kind of explanations.

In future work, we will develop an algorithm to approximate decision boundaries and to identify reasons for misclassifications across clusters of the feature space. We will improve the scalability of our tool $\mathrm{S_PEX_{PL}AI_N}$ by using optimizations tailored to NNs, and aim to handle other NN structures, such as convolutional NNs. Finally, we will apply our method to analyze how decisions evolve across the hidden layers of the network.

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